

Lab 06: Leaf Procedure in RISC-V

Objectives: Leaf Procedure in RISC-V

After completing this lab, you will:

- Implement a set of procedures;
- Ask the user to choose a procedure to execute;
- Ask the user to enter the arguments for the chosen procedure.

Interacting with the user

RARS allows to interact with the user using dialogue messages. These dialogue messages are also used to ask the user to enter some data (int, float, or string). The table given in the previous lab illustrates how to invoke such services. Execute the following RISC-V code on RARS to see these dialogue messages.

```
1 .data
2 msg1: .asciz "How old are you?"
3 Error_msg: .asciz "Age can't be negative!"
4 Warning_msg: .asciz "Are you a student?"
5 Ok_msg: .asciz "Start with the assignment!"
6 .text
7 li a7, 51
8 la a0, msg1
9 ecall
10 li t0, 18
11 li a7, 55
12 blt a0, x0, Error
13 blt a0, t0, Warning
14 la a0, Ok_msg
15 li a1, 1
16 ecall
17 beq x0, x0, End
18 beq x0, x0, End
19 Warning:
20 la a0, Warning_msg
21 li a1, 2
22 ecall
23 beq x0, x0, End
24 Error:
25 la a0, Error_msg
26 li a1, 0
27 ecall
```

Listing 1: Dialogue message example

Assignment

Write a procedure to convert temperature from Celsius to Fahrenheit and vice-versa. Write another procedure that converts numbers from decimal to binary and vice-versa.

1. First, ask the user to choose a procedure to execute;
2. After that, ask the user to choose a conversion direction (decimal → binary);
3. Then, ask the user to enter the arguments for the chosen procedure;
4. Show the result in a dialogue message;
5. Finally, ask the user if he/she wants to do that again. If so, restart the process.

Try to do it step by step.